MILITARY SPECIFICATION

PAINT (TOPCOAT), VINYL CHLORIDE-ACETATE COPOLYMER, HIGH-BUILD (FOR STEEL AND MASONRY)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a vinyl paint topcoat system for use on exterior steel and masonry surfaces treated with a primer coating conforming to MIL-P-28641. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-P-320	- Pigment, Aluminum, Powder and Paste, for paint.
PPP-P-1892	- Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

MILITARY

MIL-P-28641	- Primer Coating, Vinyl Chloride-Acetate Copolymer, High-Build (For Steel and Masonry).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Facilities Engineering Command, Code 0432, 200 Stovall Street, Alexandria, VA 22332, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

FED-STD-595 - Colors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extend specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B117 - Salt Spray (Fog) Testing.

D476 - Titanium Dioxide Pigments.

D659 - Evaluating Degree of Chalking of Exterior Paints.

D1475 - Density of Paint, Varnish, Lacquer, and Related Products.

D2621 - Infrared Identification of Vehicle Solids from Solvent-Type Paints.

D3335 - Low Concentrations of Lead and Cadmium in Paint by Atomic Absorption Spectroscopy.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

STEEL STRUCTURES PAINTING COUNCIL

SSPC-SP10 - Near White Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Solvents used shall conform to the following requirements:

a. Solvents with an olefinic or cyclo-olefinic type of unsaturation: 5 percent maximum.

b. A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethyl benzene: 8 percent maximum.

c. A combination of ethyl benzene, ketones having branched structures, or toluene: 20 percent maximum.

d. Total of a + b + c: 20 percent maximum.

3.1.1 Pigments. The pigment for white, light tints and gray shall be composed primarily of chalk-resistant rutile titanium dioxide conforming to ASTM D476, Type III or Type IV for white paints and Type IV only for tinted paints. Tinting pigments shall be lightfast and alkali resistant. Latitude is given in the selection of extender pigments provided other requirements of this specification are met. The prime pigment for olive drab and brown shall consist of synthetic iron oxides. Aluminum pigments shall conform to TT-P-320, Type II, Class 1. All pigments shall be compatible with the vehicle, and shall be nontoxic. All pigments shall provide complete hiding at the film thickness specified when tested as specified in 4.4.9.

3.1.2 Vehicle. The vehicle shall consist entirely of a solution of vinyl chloride-acetate copolymer produced by a solution polymerization process containing 85 to 88 percent vinyl chloride and 12 to 15 percent vinyl acetate by weight (See Table II). Other necessary additives such as dispersants, preservatives, etc, may be present, provided the paint meets all the requirements of the specification.

3.2 Quantitative Requirements. The paint shall conform to the quantitative requirements of Table I.

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TABLE 1.

Quantitative Requirements.

Characteristics	Requirements Minimum	Maximum
Total solids, percent by weight of paint		
White and light tints	36	--
Aluminum	30	--
Pigment, percent by weight of paint		
White and light tints	14	--
Aluminum	8	--
Total vehicle solids, percent by weight of paint		
White and light tints	20	--
Aluminum	22	--
Weight per gallon of paint, pounds		
White and light tints	9	--
Aluminum	8.5	--
Viscosity, Krebs-Stormer, shearing rate 200 r/min		
Grams	450	600
Equivalent Krebs Units (K.U.)	110	120
Dry hard time, hours	--	2
Lead, metal, percent by weight of nonvolatile matter	--	0.06

3.3 Qualitative requirements.

3.3.1 Condition in container. The paint in a freshly opened container shall show no curdling, livering, caking, or skinning, and no pigment settling which cannot be redispersed to a homogeneous mixture by stirring with a paddle, when tested as specified in 4.4.1.

3.3.2 Storage stability. The paint, after 30 days storage in a partially filled container, shall show no skinning, livering, or caking, and no pigment settling which cannot be redispersed to a homogeneous mixture by stirring with a paddle, when tested as specified in 4.4.

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3.3.3 Color. The color of the paint shall conform to FED-STD-595, color number as specified (see 6.2).

3.3.4 Working properties. The paint, when applied by airless spray, shall show no lapping, sagging, or other imperfections when tested as specified in 4.4.2.

3.3.5 System performance.

3.3.5.1 Water resistance. The coating system shall show no softening, blistering, peeling, loss of adhesion, or other film irregularities. after 21 days immersion in distilled water, when tested as specified in 4.4.3.

3.3.5.2 Salt-water resistance. The coating system shall show no softening, blistering, loss of adhesion, or other film irregularities, after 21 days of immersion in a 5 percent aqueous solution of sodium chloride when tested as specified in 4.4.4.

3.3.5.3 Salt-spray resistance. The coating system shall show no softening, blistering, peeling, embrittlement, or corrosion extending beyond 1/8 inch of the score line of the test panel, after 300 hours exposure to salt spray when tested as specified in 4.4.5.

3.3.5.4 Accelerated weathering resistance. The coating system shall show no film failure except for no more than moderate chalking, after 300 hours accelerated weathering. When tested as specified in 4.4.7 moderate chalking shall be defined as No. 6.

3.5.5.5 Chemical resistance. The coating system shall show no disintegration, blistering, underfilm corrosion, or loss of adhesion when tested as specified in 4.4.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of preparation for delivery (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling and inspection. Sampling and inspection shall be in accordance with Methods 1011, 1021, and 1031 of FED-STD-141 as applicable.

4.4 Tests. Samples selected in accordance wit 4.3.1 shall be tested as specified in Table II. Nonconformance to the requirements of the applicable test methods or failure to pass any test shall be cause for rejection of the lot which the sample represents.

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TABLE II. Tests and Methods

Characteristics	Requirement Reference	Test Method	Paragraph Reference
Hiding power	3.1.1	4121[1/]	4.4.9
Condition in container	3.3.1	3011[1/]	4.4.1
Storage stability	3.3.2	3022[1/]	
Color	3.3.3	4250[1/]	
Working properties	3.3.4		4.4.2
Water resistance	3.3.5.1	6011[1/]	4.4.3
Salt water resistance	3.3.5.2		4.4.4
Salt spray resistance	3.3.5.3	B117[2/]	4.4.5
Accelerated weathering resistance	3.3.5.4	D659[1/]	4.4.6
Chemical resistance	3.3.5.5		4.4.7
Pigment	Table I		4.4.8
Total solids	Table I		4.4.8
Nonvolatile organic	Table I		4.4.8
Weight per gallon	Table I	D1472[2/]	
Dry hard time	Table I	4061[1/]	
Lead content	Table I	D3335[2/]	
Identification of resin	3.1.2	D2621[2/]	

1/ FED-STD-141
2/ ASTM

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4.4.1 Condition in container. Examine the paint as received in accordance with Method 3011 of Fed-Std-141 for compliance with 3.3.1.

4.4.2 Working properties. Determine the spraying properties using airless spray equipment at gun pressures of over 14,000 kPa (2000 psig). For application, do not thin the paint. For the substrate, use abrasive blasted cold rolled steel test panels. The abrasive blasted surface is a near-white blast cleaned surface as defined by SSPC-SP10. After coating, evaluate for compliance with 3.3.4.

4.4.3 Water resistance.

4.3.3.1 Panel preparation. Test panels shall be approximately 100 by 200 mm by 1.625 mm (approximately 4 by 8 inches by 0.064-inch), cold rolled steel[2/] abrasive blasted in accordance with SSPC-SP10, and solvent cleaned with mineral spirits. The panels shall be coated with one coat of primer conforming to MIL-P-28641, of 0.3 mm (0.012 inch) wet film thickness, using an adjustable film casting knife[1/]. After drying two hours, apply a coat of the paint of 0.3 mm (0.012 inch) wet film thickness using an adjustable film casting knife. The cut edges of the panels shall be protected with a suitable coating stable under the conditions of the test, such as carnauba wax. The coating shall be dried one week before testing for compliance with 3.3.5.1.

4.4.3.2 Exposure testing. Place the panels in a beaker of distilled water as described in Method 6011 of FED-STD-141. The panels shall be immersed 21 days at 23 ± 1°C (73 ± 2°F). After exposure, allow panels to dry for one hour and examine the exposed panels for compliance with 3.3.5.1.

4.4.4 Salt-water resistance. Panel preparation for this test is that described in 4.4.3.1. Exposure testing is that described in 4.4.3.2 except that the panels are immersed in a 5 percent aqueous sodium chloride solution, After exposure, examine the exposed panels for compliance with 3.3.5.2

4.4.5 Salt-spray resistance. Panel preparation for this test is that described in 4.4.3.1 except that a 50 mm (2-inch) "X" shall be scribed to the metal on the lower half of the panel before exposure testing. Place the panels in an exposure cabinet as described in ASTM B117 using a 5 percent sodium chloride solution. Expose the panels for 300 hours. Observe the exposed panels for compliance with 3.3.5.3.

4.4.6 Accelerated weathering resistance. Panel preparation for this test is that described in 4.4.3.1 Subject the panels to accelerated weathering using a single carbon arc as light source, in accordance with Method 6152 of FED-STD-141 for 300 hours. After exposure, allow the panel to dry and test for chalking as described in ASTM D659 for compliance with 3.3.5.4.

1/ Available from Gardner Laboratory, Inc., 5512 Landy Lane, Bethesda, MD 20014
2/ Available from IMCO Laboratories, Inc., 1800 Broadway, Buffalo, NY 14212

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4.4.7 Chemical resistance (Except Aluminum Pigmented Paints). Panel preparation for this test is that described in 4.4.3.1. Glass cylinders 25.4 mm (1 inch) I.D. and 12.7 mm (1/2 inch) height and sealed to the coated surface with carnauba wax. A 10 percent aqueous sodium hydroxide solution is placed in a sealed glass cylinder and a watch glass is used to cover the cylinder. Add 10 percent sulfuric acid solution to another cylinder and a watch glass is used to cover the cylinder. The panels are exposed one week. The sodium hydroxide and sulfuric acid solutions are rinsed off with water and the panels allowed to dry. After the panels have been exposed, examine the panels for compliance with 3.3.5.5.

4.4.8 Pigment and nonvolatile organic material.

4.4.8.1 Method. Spread approximately 5 grams of well-mixed paint sample thinly over the surface of a tared petri dish of borasilicate glass, approximately 100 mm in diameter, and obtain the exact weight. Dry in a vented regular oven at 105°C (224°F) to constant weight. (This requires overnight drying.) After obtaining the weight of the total nonvolatile material (total solids), ash in a muffle furance, raising the temperature gradually to 450-500°C (840-930°F). Allow 2 hours for complete combustion of the organic material. The furnace door should be opened carefully when removing the sample as the remaining pigment is very light and easily scattered. Place the sample in desiccator and cool to room temperature. Obtain the weight of the ashed material.

4.4.8.1 Calculations.

Weight of total nonvolatile material - weight of ashed material = weight of nonvolatile organic material

$$\frac{\text{weight of ash}}{\text{total weight of sample}} \times 100 = \text{pigment, percent by weight of paint}$$

$$\frac{\text{wt. of nonvolatile organic material}}{\text{total weight of sample}} \times 100 = \text{nonvolatile organic material, percent by wt. of paint}$$

4.4.9 Dry opacity. The dry opacity shall be determined in accordance with Method 4121, Procedure B, of FED-STD-141, with a 0.3 mm (0.012 inch) wet film thickness to determine compliance with 3.1.1.

4.5 Inspection of preparation for delivery. The packaging, pacing, and marking of the paint shall be inspected to determine conformance to the requirements of Section 5 of this specification.

5. PACKAGING

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging

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shall be A, B, or C, and the level of packing shall be A, B, or C, as specified (see 6.2). The paint shall be furnished in 1 liter or 4 liter (1-quart or 1-gallon) multiple friction plug metal cans, in 20 liter (5-gallon) lug cover steel pails, or in 1 750 liter (55-gallon) full removable cover metal drums, as specified (see 6.2).

5.1.1 Additional marking. In addition to the marking requirements specified in 5.1, each container shall be marked or labeled with the type of solvent for thinning purposes recommended by the manufacturer, plus thinning instruction. The following additional marking shall be on each container.

INSTRUCTIONS FOR USE

Coatings applied by airless spray will allow application of required film thickness in two coats: primer and finish coat. However, it may be desirable to specify a different color primer to insure adequate coverage of the topcoat. In either case, primer coverage should be a maximum of 2.5 m^2/L (100 square feet per gallon). The primer should be allowed to dry a minimum of two hours. The finish coat should be applied at a rate 2 m^2/L (80 square feet per gallon). Total film build should be a minimum of 230 um (9.0 mils and a maximum of 300 um (12.0 mils).

A final drying time of 72 hours shall be allowed before placing the cooling system in service. In extremely cold weather a longer dry time will insure that all solvent has left the film before coating is put in service.

No thinning is required. Use a 26:1 (or higher) ratio airless spray unit with an input pressure of 700 kPa (100 psig).

Particular attention should be given to the thorough coating of welds, seams, corners, rivets, and similar surface irregularities. All such areas should be double lapped.

WARNING: For work in confined areas, a mask approved for protection against solvent vapors should be provided for the operator. In addition the work area should have an exhaust fan of sufficient capacity to keep solvent vapors below one percent by volume of air.

6. NOTES

6.1 Intended use. The vinyl coating system possesses good resistance to exterior exposure conditions and is also resistant to a wide range of chemicals. The coating system is intended to be used on properly prepared abrasive blasted (near-white) steel and concrete. The aluminum pigmented finish coat is recommended for immersion service. This high-build vinyl coating system is intended to serve in the same application covered by the Department of Interior, Bureau of Reclamation, Specification VR-6.

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6.2 Ordering data.

6.2.1 Procurment requirements. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Color required (see 3.3.3 and 6.4).

c. Level of packaging and level of packing required (see 5.1).

d. Quantity and size of container required (see 5.1).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9 (n) of the Armed Services Procurement Regulation, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements LIst (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9 (n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DD 1664
4.3.1	Certificate of Compliance	DI-E-2121

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

6.4 The various coatings are designed for use together as a protective coating system for steel or concrete surfaces and it is probable that coatings from various manufacturers will not be suitable for use with each other. Therefore, the activity concerned should purchase the complete system from one manufacturer and should specify in the invitation for bids that the primer and topcoat be furnished in contrasting colors; for example, with primer, blue-gray body, and white seal coat to facilitate coverage in application.

6.5 Resin characteristics. The coating procured under this specification should be comparable to the following composition, however, the Government assumes no responsibility for the acceptance of a product claimed to be manufactured under an identical composition:

PAINT (TOPCOAT)

Satisfactory coatings have been manufactured using the following formulations:

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HIGH BUILD ALUMINUM VINYL PAINT

	Pounds per 100 gallons	Gallons per 100 gallons	Percent by weight
Cellosolve Acetate[1/]	423.9	52.4	48.9
Methyl Ethyl Ketone[1/]	136.3	20.3	15.7
Xylene	40.3	5.6	4.6
BAKELITE VYHH Resin[1/]	171.4	15.1	19.8
Diisodecyl Phthalate	25.7	3.2	3.0
Aluminum Powder[2/]	46.5	2.2	5.4
Thixotropic Agent[3/]	22.9	1.2	2.6
	867.0	100.0	100.0

Suppliers

1/ Union Carbide Corporation, Chemicals and Plastics
2/ "Aluminum Powder 408", Alcoa Co.
3/ "Cab-O-Sil M-5", Cabot Corp.

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HIGH BUILD WHITE VINYL PAINT

	Pounds per 100 gallons	Gallons per 100 gallons	Percent by weight
Cellosolve Acetate 1/	418.3	51.6	45.2
Methyl Ethyl Ketone 1/	134.5	20.0	14.5
Xylene	39.7	5.5	4.3
BAKELITE VYHH Resin 1/	169.0	15.0	18.2
Diisodecyl Phthalate	25.4	3.1	2.7
Dispersant 2/	2.1	0.3	0.2
Titanium Dioxide 3/	120.2	3.5	13.0
Thixotropic Agent 4/	17.9	1.0	1.9
	927.1	100.0	100.0

Suppliers

1/ Union Carbide Corp., Chemicals and Plastics
2/ "Nuosperse 657", Tenneco, Inc.
3/ "Ti Pure R-966". E. I. DuPont
4/ "Cab-O-Sil M-5", Cabot Corp.

6.6 Unit of purchase. The paint shall be purchased by volume, the unit being one U. S. gallon of 231 cubic inches (approximately 3.7 liters) at 20°C (68°) until SI Packages are available.

Preparing Activity

Navy - YD

(Project Number 8010-N154)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-P-28642 - PAINT (TOPCOAT), VINYL CHLORIDE-ACETATE COPOLYMER, HIGH-BUILD (FOR STEEL AND MASONRY)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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